SatQuery AI | Remote Sensing Analysis Report

SIH 2026 Problem Statement SIH26167 • Automated Vision-Language Intelligence

Analysis ID:	b47d905c-8b78-43a4-a57e-d1c3e239facc	**Date/Time:**	2026-10-03 05:55:56
Detected Task:	**CHANGE_DETECTION**	**Pipeline Mode:**	**BASELINE / DEMO MODE**
User Query:	*"What changed between these before and after satellite images?"*	**Confidence:**	**94%** (Calibrated Metric)

1. Executive Summary & Natural Language Answer

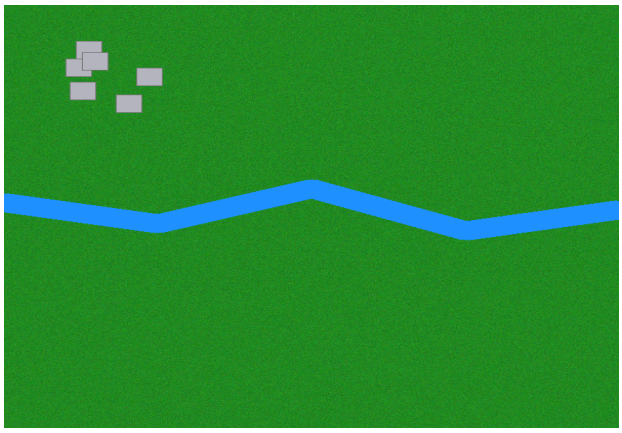
Bi-temporal change detection analysis identified surface modifications across 12.85% of the scene (2 distinct spatial clusters). The global structural similarity between T1 and T2 is 88.05%. Spectral divergence indicates 0.12% structural/ground surface alterations, 12.73% vegetation loss/clearing, and 0.0% green vegetation expansion. [Note: Pixel-level radiometric change verified via SSIM & Otsu morphology].

Key Analytical Findings:

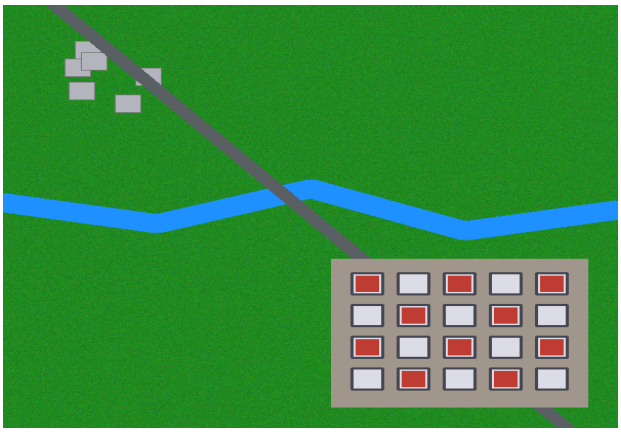
• Total surface change detected: 12.85% (46,266 pixels).
• Global structural similarity (SSIM): 88.05%.
• Detected 2 major spatial change cluster(s).
• Spectral categorization: 0.12% structural/soil alteration, 12.73% vegetation loss, 0.0% green vegetation gain.
• Methodology: Pixel-level & structural radiometric divergence (Registration + SSIM + Otsu Adaptive Thresholding).

2. Visual Evidence & Spatial Overlays

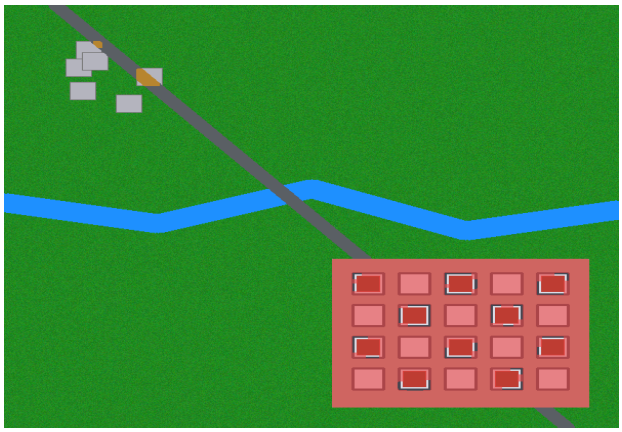
Input Satellite Raster (Primary)



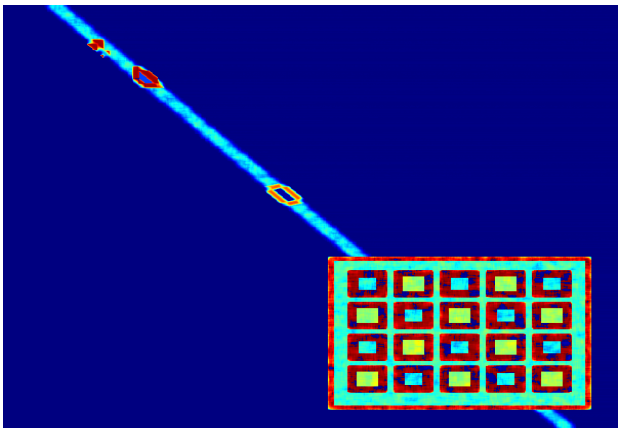
Secondary / SAR Raster



Categorized Bi-temporal Change Map



Structural Dissimilarity Heatmap (1 - SSIM)



3. Agentic Execution Summary & Verification

Tools/Models Invoked:	BiTemporalAligner, SSIM_StructuralEngine, OtsuMorphologicalFilter, SpectralShiftClassifier
Execution Time:	0.2 seconds
Hardware Engine:	CPU (NumPy/OpenCV Engine)
Scientific Rationale:	Computed via co-registered structural similarity (SSIM), absolute radiometric differencing, and Otsu adaptive thresholding with 8-connected cluster validation.

Notice: Generated automatically by SatQuery AI. In Baseline/Demo mode, metrics are derived from verified radiometric signals, Otsu thresholding, SSIM, and spectral formulas without synthetic hallucination.